

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: GLUE | Context: Luxembourgish Public Administration Civil Servants — GLUE Assessment

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GLUE's task ontology consists of nine English sentence-understanding tasks — sentiment, paraphrase, entailment, QA, acceptability, similarity [\[Q3, Q18\]](#) — with no administrative-domain categories. The deployment requires routing categories that GLUE does not contain: cross-border worker (frontalier) classification, national vs. communal (État vs. commune) competency routing, housing urgency, social security, immigration, taxation, and public-transport queries (deployment_specific_categories in YAML). The diagnostic set's 'World Knowledge' [\[Q195\]](#) and 'Common Sense' [\[Q196\]](#) categories are general English-language reasoning probes with no jurisdictional specificity. IO is marked HIGH priority and the gap is total — the dataset analysis confirms zero administrative-domain content across 409 sampled examples.

ID	Evidence
Q3	"GLUE is a collection of NLU tasks including question answering, sentiment analysis, and textual entailment, and an associated online platform for model evaluation, comparison, and analysis." (p.1)
Q18	"GLUE is centered on nine English sentence understanding tasks, which cover a broad range of domains, data quantities, and difficulties." (p.3)
Q195	"World Knowledge In this category we focus on knowledge that can clearly be expressed as facts, as well as broader and less common geographical, legal, political, technical, or cultural knowledge." (p.20)
Q196	"Common Sense In this category we focus on knowledge that is more difficult to express as facts and that we expect to be possessed by most people independent of cultural or educational background." (p.20)

Strengths

- The multi-task structure with per-task scoring plus macro-average [\[Q15, Q59, Q60\]](#) offers a methodological template for designing a Luxembourgish administrative benchmark suite that combines NER, intent classification, sentiment, and routing tasks with separate metrics.
- The diagnostic set's minimal-pair construction methodology [\[Q65, Q66, Q68\]](#) tagged for fine-grained linguistic phenomena [\[Q135, Q136\]](#) is a transferable design pattern for probing model behaviour on negation, quantifier scope, and coreference in administrative text.

ID	Evidence
Q15	"In summary, we offer: (i) A suite of nine sentence or sentence-pair NLU tasks, built on established annotated datasets and selected to cover a diverse range of text genres, dataset sizes, and degrees of difficulty. (ii) An online evaluation platform and leaderboard, based primarily on privately-held test data. The platform is model-agnostic, and can evaluate any method capable of producing results on all nine tasks. (iii) An expert-constructed diagnostic evaluation dataset. (iv) Baseline results for several major existing approaches to sentence representation learning." (p.2)
Q59	"The benchmark site shows per-task scores and a macro-average of those scores to determine a system's position on the leaderboard." (p.5)
Q60	"For tasks with multiple metrics (e.g., accuracy and F1), we use an unweighted average of the metrics as the score for the task when computing the overall macro-average." (p.5)
Q65	"We construct each example by locating a sentence that can be easily made to demonstrate a target phenomenon, and editing it in two ways to produce an appropriate sentence pair." (p.6)
Q66	"We make minimal modifications so as to maintain high lexical and structural overlap within each sentence pair and limit superficial cues." (p.6)

Q68	"Where possible, we produce several pairs with different labels for a single source sentence, to have minimal sets of sentence pairs that are lexically and structurally very similar but correspond to different entailment relationships." (p.6)
Q135	"To make this kind of analysis feasible, we first identify four broad categories of phenomena: Lexical Semantics, Predicate-Argument Structure, Logic, and Knowledge." (p.15)
Q136	"However, since these categories are vague, we divide each into a larger set of fine-grained subcategories." (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include: cross-border worker / frontali  r vs. resident classification, national (  tat) vs. communal competency routing across 100 communes [WEB-16, WEB-17, WEB-18], housing/cost-of-living urgency, social security, taxation (resident vs. non-resident), immigration, public transport (including the free-transit policy [WEB-20, WEB-21]), civil registration, EU-institutional employment, and local planning. None of these are present in GLUE.

IO-2: GLUE's taxonomy omits every regionally-relevant administrative category. The closest approximation is MNLI's inclusion of 'government reports' as one of ten premise sources [Q38], but this is a stylistic source, not an ontological category, and no government-administrative example appears in the 409-example dataset sample.

IO-3: GLUE includes categories irrelevant to the target context: English linguistic acceptability judgments from linguistics journals [Q21], movie-review sentiment [Q26], image-caption semantic similarity (STS-B) [Q35], Winograd pronoun coreference [Q50], and English textual entailment from news/Wikipedia [Q46, Q48]. The dataset analysis (DATASET-D6, DATASET-D8, DATASET-D40) confirms CoLA tests English-specific morphosyntactic phenomena with no transfer value to Luxembourgish/French/German administrative text.

IO-4: Content-validity gaps are total: the deployment's required test-case categories (administrative routing, frontali  r status,   tat/commune competency) have zero coverage. Construct-irrelevant variance is also high — performance on English movie-review sentiment, Winograd schemas, and image-caption similarity will not predict performance on Luxembourgish administrative routing.

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Q3	"GLUE is a collection of NLU tasks including question answering, sentiment analysis, and textual entailment, and an associated online platform for model evaluation, comparison, and analysis." (p.1)
Q18	"GLUE is centered on nine English sentence understanding tasks, which cover a broad range of domains, data quantities, and difficulties." (p.3)
Q21	"The Corpus of Linguistic Acceptability (Warstadt et al., 2018) consists of English acceptability judgments drawn from books and journal articles on linguistic theory." (p.3)
Q26	"The Stanford Sentiment Treebank (Socher et al., 2013) consists of sentences from movie reviews and human annotations of their sentiment." (p.3)
Q35	"The Semantic Textual Similarity Benchmark (Cer et al., 2017) is a collection of sentence pairs drawn from news headlines, video and image captions, and natural language inference data. Each pair is human-annotated with a similarity score from 1 to 5; the task is to predict these scores. Follow common practice, we evaluate using Pearson and Spearman correlation coefficients." (p.3)
Q38	"The premise sentences are gathered from ten different sources, including transcribed speech, fiction, and government reports." (p.4)
Q46	"The Recognizing Textual Entailment (RTE) datasets come from a series of annual textual entailment challenges." (p.4)
Q48	"Examples are constructed based on news and Wikipedia text." (p.4)

Q50	"The Winograd Schema Challenge (Levesque et al., 2011) is a reading comprehension task in which a system must read a sentence with a pronoun and select the referent of that pronoun from a list of choices." (p.4)
D6	The fact that no candidate was elected shows that he was inadequate. — English linguistics acceptability judgment
D8	They drank the pub. — Idiomatic English usage; grammatical acceptability judgment
D40	The children eat all chocolate. — English determiner usage; grammaticality judgment
WEB-10	ItzGLUE provides Luxembourgish sentiment (positive/neutral/negative) but not in administrative register (https://arxiv.org/abs/2604.17976)
WEB-16	100 communes as of 1 September 2023 — establishes communal-routing label space requirement absent from GLUE (https://elections.public.lu/en/fusion-communes.html)
WEB-17	https://gouvernement.lu/en/actualites/toutes_actualites/communiqués/2023/02-fevrier/10-100-communes.html
WEB-18	https://en.wikipedia.org/wiki/Communes_of_Luxembourg
WEB-19	Loi communale 1988 defines autonomous communal competencies — establishes the legal taxonomy GLUE does not cover (https://www.sng-wofi.org/country_profiles/luxembourg.html)
WEB-20	Free public transit since 29 February 2020 — domain-specific topic absent from GLUE (https://luxembourg.public.lu/en/living/mobility/public-transport.html)
WEB-21	https://transports.public.lu/en/plus/faq/gratuite-transports-publics.html

Information Gaps

- The deployment's full required topic taxonomy is itself not fully verified; the YAML notes that ranking categories by correspondence volume requires CTIE/ministerial access that was not available.

Requires Expert Verification

- Final administrative topic taxonomy and priority ordering should be elicited from CTIE and ministry stakeholders before any custom benchmark construction.

Remediation

Gap 1: GLUE's task ontology contains no administrative-routing categories, no national-vs-communal competency distinction, and no cross-border-worker classification.

Recommendation: Construct a Luxembourg-specific topic and intent taxonomy via stakeholder elicitation with CTIE and ministry staff, covering: État/commune routing across 100 communes, frontalier-vs-resident status, housing urgency, social security, taxation, immigration, public transport, civil registration, and EU-institutional matters. Use ItzGLUE's slot-intent design [WEB-10] as a structural template but rebuild the label space for administrative content.

ID	Evidence
WEB-10	ItzGLUE provides Luxembourgish sentiment (positive/neutral/negative) but not in administrative register (https://arxiv.org/abs/2604.17976)